Vasiliki Vasileiou

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VasilikiVas • Vasiliki-Vasileiou

Research Interests

My main areas of interest are Computer Vision (3D Reconstruction, Rendering, Segmentation, Detection, Temporal Segmentation on videos, Autonomous Driving) and Robotics (Applications to Computer Vision, Visual Servoing, Grasping, Manipulation).

Education

National Technical University of Athens (NTUA), Greece Sept. 2023 – Now
PhD Candidate in Geometric Computer Vision

- Current GPA: 9.6/10
- PhD Fellow at Archimedes Research Unit and HERON Robotics CoE, Athena RC
- Advisors: Prof. Kostas Daniilidis (UPenn), Prof. Petros Maragos (NTUA)

University of Amsterdam (UvA), The Netherlands Sept. 2021 – Jul. 2023
MSc in Artificial Intelligence (2 years; 120 ECTS)

- Final GPA: 7.55/10
- Thesis: "Language Driven 3D Scene Understanding"
- Advisors: Prof. Martin R. Oswald (UvA), Dr. Silvan Weder (ETH)

National Technical University of Athens (NTUA), Greece Sept. 2015 – Feb. 2021
BSc & MEng in Electrical and Computer Engineering (5-year joint degree; 300 ECTS)

- Final GPA: 8.24/10 (estimated at top 12-15%)
- Concentration: Signal, Control Systems and Robotics & Software Systems & Computing Systems
- Thesis: "Deep Neural Network Architectures for human action/gesture recognition and temporal segmentation in videos"
- Advisor: Prof. Petros Maragos (NTUA)

Coursework

Undergraduate (among others): Computer Vision, Intelligent Systems, Pattern Recognition, Robotics, Speech and Language Processing, Human-Computer Interaction

Graduate (among others): Computer Vision, Deep Learning, Machine Learning, Data mining, Natural Language Processing, Deep Learning for Natural Language Processing, Machine Perception

Professional and research experience

PhD Candidate Jul. 2026 – Now
HERON Robotics CoE - Athena RC

- Working with Neural Radiance Fields, Projective Geometry, 3D Geometry in Computer Vision, Language models

PhD Candidate Sep. 2023 – Jul. 2026
Archimedes Research Unit - Athena RC

- Working with Neural Radiance Fields, Projective Geometry, 3D Geometry in Computer Vision, Language models

Visiting Associate Jan. 2024 – Mar. 2024
GRASP Lab - University of Pennsylvania (UPenn)

- Visiting my PhD Advisor
- Attending the Machine Perception Course
- Working on my PhD research projects

AI Research Intern Nov. 2022 – Jul. 2023
ATLAS Lab (UvA-TomTom)

- Working on my master's thesis on the topic Language Driven 3D Scene Understanding
- Working with Language models, Pointclouds, 3D learning with 2D supervision, Open-vocabulary tasks

Teaching Assistant Jun. 2022 – May 2023

University of Amsterdam & Vrije Universiteit Amsterdam

- Responsible for:
  - The labs of the 1st year master's course computer vision 1
  - The work colleges of the 1st year bachelor's course Data structures and Algorithms
  - The labs of the 2nd year master's course Socially Intelligent Robotics
  - The labs of the 1st year bachelor's course Human-Computer Interaction

Undergraduate Research Assistant Mar. 2020 – Jul. 2021

Computer Vision, Speech Communication & Signal Processing Group, NTUA

- Working on human action/gesture recognition and temporal segmentation
- Working with 2D/3D skeletons, RGB/Depth/Optical Flow features, LSTM/3DCNN/TRN/I3D models and other Machine Learning techniques

Publications

[1] **Vasiliki Vasileiou**, Panagiotis P. Filntisis, Petros Maragos, Kostas Daniilidis "ReVA-HPE: Relative 6DoF Head Pose Estimation with Synthetic Pose Supervision and Personalized Virtual Anchors", WACV 2027

[2] **Vasiliki Vasileiou**, Panagiotis P. Filntisis, Petros Maragos, Kostas Daniilidis "VGGT-HPE: Reframing Head Pose Estimation as Relative Pose Prediction", CVPR Workshop 2026 (oral)

[3] **Vasiliki Vasileiou**, Chase Myers van de Geijn, Victor Kyriacou, Irene Papadopoulou, Reproducibility study for "Explaining in Style: Training a GAN to explain a classifier in StyleSpace", ReScience C 2022 (Journal Showcase of NeurIPS 2022)

[4] **Vasiliki Vasileiou**, Nikolaos Kardaris, Petros Maragos, "Exploring Temporal Context and Human Movement Dynamics for Online Action Detection in Videos", EUSIPCO 2021 (oral)

Strengths and Skills

Programming: Python, C/C++, MATLAB, Java, Prolog, OCaml
Machine Learning Frameworks: PyTorch, Tensorflow Keras, Sklearn
Libraries: OpenCV, Open3D, COLMAP, PyTorch3D, Pandas, NumPy, Matplotlib
Extracurricular: Tennis, Chess

Honors and Awards

Master Studies Scholarship | ***Eugenides Foundation***
A scholarship to a graduate of the School of Electrical Engineering & Computer Engineering of NTUA, for postgraduate studies in the subjects of engineering at a university abroad, full time & attendance with a physical presence.

A Great Moment for Education | ***Ministry of Education & Eurobank EFG***
€800 award for succeeding the highest Nationwide University Entrance Examination Score at my Senior High School

Volunteering

Assistant at 2nd High School of Preveza: Solving problems and questions of my classmates based on Mathematics and Programming after school hours once per week, Preveza 2013 - 2015

The Smile of Child: Creativity Lab member, Athens 2020

Athens Marathon the Authentic: Marathon course support stations, Athens 2019

Athens Science Festival: School Host, Athens 2019

Languages

English (Fluent), **French** (Upper-intermediate), **Greek** (Native)

References

Kostas Daniilidis (Upenn - PhD Advisor)
Petros Maragos (NTUA - PhD Advisor)
Martin R. Oswald (UvA - MSc Thesis Advisor)